

The **Rejection Method** (also known as the acceptance-rejection method) is a versatile technique for generating random variables from a target distribution when the **Inverse Transformation Method** is impractical due to a mathematically complex cumulative distribution function (CDF) 1. While the Inverse Transformation Method requires a simple, invertible form of the CDF, the Rejection Method only requires that the target probability density function (PDF), $f(x)$, is available 1.

Core Mechanism and Procedure

The technique involves using a "proposal" or "instrumental" distribution with a density function $g(x)$ from which it is easy to generate samples 2. The sources outline a three-step algorithm to achieve this:

1. **Generate a candidate:** Produce a random variable Y from the proposal density g 2.
2. **Generate a uniform test:** Produce a random number U from a **Standard Uniform distribution** ($\text{Uniform}(0, 1)$) 2.
3. **The Accept/Reject Step:** Check if $U \leq \frac{f(Y)}{cg(Y)}$, where c is a constant such that $\frac{f(y)}{g(y)} \leq c$ for all y 2. If this condition is met, set $X = Y$; otherwise, reject Y and return to the first step 2.

The mathematical validity of this method rests on the fact that the conditional distribution of Y , given that it satisfies the rejection criteria, is exactly the target distribution F 3.

Efficiency and Performance

The efficiency of the Rejection Method is determined by the constant c . The sources explain that the number of iterations required to successfully generate one random variable follows a **geometric distribution** with a success probability of $p = 1/c$ 4.

- **Expected Trials:** The expected number of attempts needed is exactly c 4.
- **Optimal Proposals:** To minimise computational cost, c should be as small as possible, which requires a proposal density $g(x)$ that closely matches the shape of the target density $f(x)$ 4, 5.

Applications and Examples

The sources provide several concrete examples of how this method is applied to common distributions:

- **Beta Distribution:** Generating a **Beta(2, 4)** variable by using a $\text{Uniform}(0, 1)$ distribution as the proposal $g(x)$ 5.
- **Normal Distribution:** Generating a **Standard Normal** variable using an **Exponential distribution** with a mean of 1 as the proposal 6.
- **Gamma Distribution:** Simulating a **Gamma(5/2, 1)** variable or a conditional Gamma distribution (e.g., $\text{Gamma}(2, 1)$ conditioned to be greater than 5) 7, 8.

Larger Context of Simulation Techniques

Within the broader framework of probabilistic modeling and simulation, the Rejection Method serves as a foundational building block for more advanced techniques:

- **MCMC and Gibbs Sampling:** In complex Bayesian modeling, the Rejection Method is sometimes nested within other algorithms. For instance, **Adaptive Rejection Sampling** can be used within **Gibbs sampling** when drawing from full conditional distributions is not straightforward 9.
- **Sequential Monte Carlo (SMC):** Advanced particle methodologies like SMC or **particle filters** can be interpreted as sophisticated acceptance-rejection samplers equipped with "interacting recycling mechanisms" 10.
- **Refinement of Monte Carlo Integration:** The method relates to **importance sampling**, where points are sampled more frequently in regions where the integrand is large to improve the accuracy of numerical results 11.

While powerful, the Rejection Method's main limitation is that it can become computationally expensive in high-dimensional spaces or when a suitable proposal distribution $g(x)$ with a small c cannot be found 1, 12.